

Airbnb Price Prediction – NYC

1. Objective

The objective of this project is to predict Airbnb listing prices in New York City using machine learning techniques based on location, room type, host activity, and review-related features. The goal is to assist hosts in setting competitive prices to maximize revenue.

2. Dataset Description

The dataset used is the Airbnb NYC Listings dataset containing 48,895 listings with 16 attributes, including neighbourhood, room type, price, reviews, availability, and host listing count.

3. Data Preprocessing

Data preprocessing steps included removal of missing values, handling of outliers using the Interquartile Range (IQR) method, and conversion of categorical variables into numerical form using one-hot encoding.

4. Exploratory Data Analysis

EDA revealed significant price variation based on room type and neighbourhood. Entire homes/apartments were priced higher compared to private or shared rooms. Manhattan emerged as the most premium neighbourhood group.

5. Models Used

Two regression models were implemented: Linear Regression as a baseline model and XGBoost Regressor as an advanced ensemble learning method.

6. Model Performance

Model	MAE	RMSE	R ²
Linear Regression	34.04	46.35	0.525
XGBoost	31.23	43.33	0.585

7. Feature Importance

Room type and neighbourhood were the most influential features in determining price. Listings in Manhattan, Midtown, and Upper East Side commanded higher prices. Private and shared rooms significantly reduced listing prices.

8. Conclusion

The XGBoost model outperformed Linear Regression and was selected as the final model. The model can effectively predict Airbnb prices and provide actionable insights for hosts to optimize their pricing strategy.

9. Future Scope

Future improvements include incorporating amenities, host response time, customer ratings, and deploying the model as a web application for real-time price prediction.